

AERO70015/96012 Mathematics 3 Problem Set #4

Problem Sheet 4

4.1 PR4_1

(a) Using the convolution theorem, show that

$$\mathcal{L}^{-1} \left\{ \frac{1}{s(s+a)} \right\} = \frac{1}{a} (1 - e^{-a t}).$$

(b) Hence obtain the solution of the o.d.e.

$$x''(t) + a x'(t) = f(t), \quad x(0) = x_0, \quad x'(0) = x_1,$$

where $f(t)$ is an arbitrary function.

Show that the steady-state value of x is given by

$$x(\infty) = \frac{1}{a} \int_0^{\infty} f(u) du + x_0 + \frac{x_1}{a}.$$

$$(a) \quad \mathcal{L}^{-1} \left\{ \frac{1}{s(s+a)} \right\} = \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} \mathcal{L}^{-1} \left\{ \frac{1}{s+a} \right\}$$

$$= \int_0^t e^{-at} \cdot 1 \, d\tau$$

$$= \left[-\frac{1}{a} e^{-a\tau} \right]_0^t$$

$$= \frac{1}{a} (1 - e^{-at})$$

(b) Apply Laplace transformations:

$$s^2 X - s x(0) - x'(0) + a [sX - x(0)] = F(s)$$

$$\therefore (s^2 + as) X - (a+s)x_0 - x_1 = F(s)$$

$$\therefore X = \frac{1}{s(s+a)} [F(s) + (s+a)x_0 + x_1]$$

$$\therefore x = \mathcal{L}^{-1} \left\{ \frac{F(s)}{s(s+a)} \right\} + \mathcal{L}^{-1} \left\{ \frac{s+a}{s(s+a)} x_0 \right\} + \mathcal{L}^{-1} \left\{ \frac{1}{s(s+a)} x_1 \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{1}{s(s+a)} \right\} \mathcal{L}^{-1} \{ F(s) \} + \mathcal{L}^{-1} \left\{ \frac{x_0}{s+a} \right\} + \mathcal{L}^{-1} \left\{ \frac{a}{s(s+a)} x_0 \right\} + \mathcal{L}^{-1} \left\{ \frac{1}{s(s+a)} x_1 \right\}$$

$$= \frac{1}{a} \int_0^t (1 - e^{-a(t-u)}) f(u) du + e^{-at} x_0 + (1 - e^{-at}) x_0 + \frac{1}{a} (1 - e^{-at}) x_1$$

$$\therefore x(\infty) = \frac{1}{a} \int_0^{\infty} f(u) du + x_0 + \frac{1}{a} x_1.$$

4.2 PR4_2

(a) Using the convolution theorem, show that if

$$y(x) = f(x) + \int_0^x g(x-u) y(u) du,$$

then $Y(s) = F(s)/(1 - G(s))$, where Y , F and G are the Laplace transforms of y , f and g respectively.

(b) Hence solve the integral equation

$$y(x) = \sin(3x) + \int_0^x \sin(x-u) y(u) du.$$

(a) Apply Laplace transformations:

$$Y(s) = F(s) + G(s)Y(s)$$

$$\therefore [1 - G(s)]Y(s) = F(s)$$

$$\therefore Y(s) = \frac{F(s)}{1 - G(s)}$$

$$\begin{aligned} \text{(b)} \quad Y(s) &= \frac{F(s)}{1 - G(s)} = \frac{\frac{3}{s^2 + 9}}{1 - \frac{1}{s^2 + 1}} \\ &= \frac{3(s^2 + 1)}{(s^2 + 9)(s^2 + 1 - 1)} \end{aligned}$$

$$\therefore Y(s) = \frac{3(s^2+1)}{s^2(s^2+9)} = \frac{\frac{1}{3}}{s^2} + \frac{\frac{8}{3}}{s^2+9}$$

$$\begin{aligned}\therefore y(x) &= \mathcal{L}^{-1}\{Y(s)\} = \frac{1}{3}\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} + \frac{8}{9}\mathcal{L}^{-1}\left\{\frac{3}{s^2+9}\right\} \\ &= \frac{1}{3}x + \frac{8}{9}\sin(3x)\end{aligned}$$

4.3 PR4_3

Find

$$\mathcal{L}^{-1}\left\{\frac{s}{(s^2 + \omega^2)^2}\right\}$$

(with ω a positive constant) by the following methods:

- Splitting into partial fractions and using the shift theorems and known transforms;
- Using the convolution theorem together with known transforms;
- Using the complex inversion formula.

$$(a) \quad \frac{s}{(s^2 + \omega^2)^2} = \frac{1}{4i\omega} \left[-\frac{1}{(s+i\omega)^2} + \frac{1}{(s-i\omega)^2} \right]$$

$$\therefore \mathcal{L}^{-1}\left\{\frac{s}{(s^2 + \omega^2)^2}\right\} = \frac{1}{4i\omega} \left\{ -\frac{1}{(s+i\omega)^2} \right\} + \frac{1}{4i\omega} \left\{ \frac{1}{(s-i\omega)^2} \right\}$$

$$= \frac{1}{4i\omega} (-e^{-i\omega t} t + e^{i\omega t} t) = \frac{2i\sin(\omega t)t}{4i\omega} = \frac{\sin(\omega t)t}{2\omega}$$

$$(b) \frac{s}{(s^2 + \omega^2)^2} = \frac{s}{(s^2 + \omega^2)} \frac{1}{(s^2 + \omega^2)} = \frac{1}{\omega} \frac{s}{(s^2 + \omega^2)} \frac{\omega}{(s^2 + \omega^2)}$$

$$\therefore \mathcal{L}^{-1} \left\{ \frac{s}{(s^2 + \omega^2)^2} \right\} = \frac{1}{\omega} \int_0^t \cos(\omega u) \sin[\omega(t-u)] du$$

$$\text{where } \sin a \cos b - \cos a \sin b = \sin(a-b)$$

$$\begin{aligned} \therefore \mathcal{L}^{-1} \left\{ \frac{s}{(s^2 + \omega^2)^2} \right\} &= \frac{1}{\omega} \int_0^t \cos(\omega u) [\sin(\omega t) \cos(\omega u) - \cos(\omega t) \sin(\omega u)] du \\ &= \frac{\sin(\omega t)}{\omega} \int_0^t \cos^2(\omega u) du - \frac{\cos(\omega t)}{\omega} \int_0^t \cos(\omega u) \sin(\omega u) du \\ &= \frac{\sin(\omega t)}{\omega} \int_0^t \frac{1}{2} [1 + \cos(2\omega u)] du - \frac{\cos(\omega t)}{\omega} \int_0^t \frac{1}{2} \sin(2\omega u) du \\ &= \frac{1}{\omega} \sin(\omega t) \left[\frac{1}{2} u + \frac{1}{4\omega} \sin(2\omega u) \right]_0^t - \frac{1}{\omega} \cos(\omega t) \left[-\frac{1}{4\omega} \cos(2\omega u) \right]_0^t \\ &= \frac{1}{2\omega} \sin(\omega t) t + \frac{1}{4\omega^2} \sin(2\omega t) \sin(\omega t) + \frac{1}{4\omega^2} [\cos(2\omega t) - 1] \cos(\omega t) \\ &= \frac{1}{2\omega} \sin(\omega t) t + \frac{1}{2\omega} \sin^2(\omega t) \cos(\omega t) + \frac{1}{2\omega^2} [\cos^2(\omega t) - 1] \cos(\omega t) \\ &= \frac{1}{2\omega} \sin(\omega t) t + \frac{1}{2\omega} [\sin^2(\omega t) + \cos^2(\omega t) - 1] \cos(\omega t) \\ &= \frac{\sin(\omega t) t}{2\omega} \end{aligned}$$

$$(c) \text{ Let } f(s)e^{st} = \frac{se^{st}}{(s^2 + \omega^2)^2} = \frac{se^{st}}{(s^2 + \omega^2)(s^2 + \omega^2)} = \frac{se^{st}}{(s+i\omega)(s-i\omega)^2}$$

$$\text{Let } p = se^{st}, q = (s^2 + \omega^2)^2$$

$$\therefore p' = (1+st)e^{st}, q' = 4s(s^2 + \omega^2), q'' = 12s^2 + 4\omega^2, q''' = 24s$$

$$\text{At } s_0 = i\omega, p(s_0) = i\omega e^{i\omega t}, p'(s_0) = (1+i\omega t)e^{i\omega t}, q(s_0) = 0, q'(s_0) = 0, q''(s_0) = -8\omega^2, q'''(s_0) = 24i\omega$$

$$\text{At } s_0 = -i\omega, p(s_0) = -i\omega e^{-i\omega t}, p'(s_0) = (1-i\omega t)e^{-i\omega t}, q(s_0) = 0, q'(s_0) = 0, q''(s_0) = -8\omega^2, q'''(s_0) = -24i\omega$$

$$\begin{aligned} \therefore \mathcal{L}^{-1} \left\{ \frac{s}{(s^2 + \omega^2)^2} \right\} &= \text{Res}_{s=i\omega} f(z)e^{st} + \text{Res}_{s=-i\omega} f(z)e^{st} \\ &= \frac{2(1+i\omega t)e^{i\omega t}}{-8\omega^2} - \frac{2i\omega e^{i\omega t}(24i\omega)}{3(64\omega^4)} + \frac{2(1-i\omega t)e^{-i\omega t}}{-8\omega^2} - \frac{2i\omega e^{-i\omega t}(24i\omega)}{3(64\omega^4)} \\ &= -\frac{e^{i\omega t} + e^{-i\omega t}}{4\omega^2} - \frac{it(e^{i\omega t} - e^{-i\omega t})}{4\omega} + \frac{e^{i\omega t} + e^{-i\omega t}}{4\omega^2} = \frac{\sin(\omega t) t}{2\omega} \end{aligned}$$

4.4 PR4_4

Solve the partial differential equation

$$\frac{\partial u}{\partial t} + x \frac{\partial u}{\partial x} = x, \quad u(x, 0) = 0, \quad u(0, t) = 0$$

for $u(x, t)$ (with $x > 0, t > 0$) by taking the Laplace transform with respect to t .

Apply Laplace transform.

$$sU - u(x, 0) + x \frac{\partial U}{\partial x} = \frac{x}{s}$$

$$\therefore \frac{\partial U}{\partial x} + \frac{s}{x} U = \frac{1}{s}$$

$$\text{let integral factor } I(x) = e^{\int \frac{s}{x} dx} = e^{s \ln x} = x^s$$

$$\therefore \frac{d}{dx} (Ux^s) = \frac{x^s}{s}$$

$$\therefore Ux^s = \int \frac{x^s}{s} dx = \frac{1}{s(s+1)} x^{s+1} + C$$

$$\therefore U = \frac{x}{s(s+1)} + Cx^{-s} = \frac{x}{s} - \frac{x}{s+1} + Cx^{-s}$$

$$\therefore u = \mathcal{L}^{-1}\{U\} = x(1 - e^{-t}) + \mathcal{L}^{-1}\{Cx^{-s}\}$$

$$\therefore u(0, t) = 0 + \mathcal{L}^{-1}\{Cx^{-s}\} = 0$$

$$\therefore \mathcal{L}^{-1}\{Cx^{-s}\} = 0$$

$$\therefore u = x(1 - e^{-t})$$

4.5 PR4_5

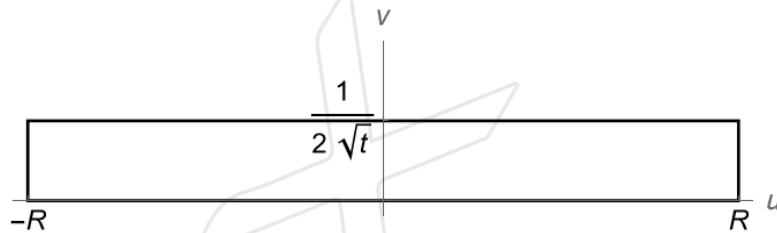
- (a) Using a keyhole Bromwich contour, with a branch cut along the negative real axis, show that the inverse Laplace transform of $F(s) = e^{-\sqrt{s}}$ is equal to

$$\frac{1}{\pi} \int_0^\infty e^{-r t} \sin(\sqrt{r}) dr.$$

(You may assume that the integrals along the outer curves tend to zero as $R \rightarrow \infty$.)

- (b) Using the substitution $r t = u^2$, show that this is equal in turn to the imaginary part of

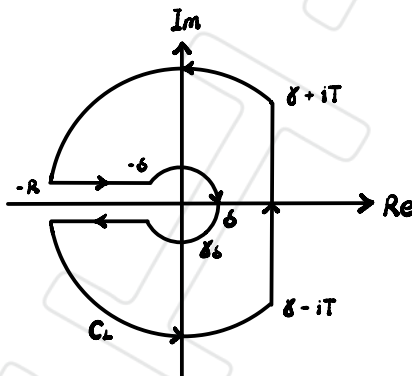
$$\frac{1}{\pi t} \int_{-\infty}^\infty u e^{-u^2} e^{i u/\sqrt{t}} du.$$



- (c)

By considering the integral of $w e^{-(w-i/[2\sqrt{t}])^2}$ around the contour shown in the figure, where $w = u + i v$, and letting R tend to infinity, express this imaginary part in terms of t , π and the known integral $\int_{-\infty}^\infty e^{-u^2} du = \sqrt{\pi}$, and hence find the inverse Laplace transform of $F(s) = e^{-\sqrt{s}}$.

(a)



$$\begin{aligned} & \oint_{C_L} e^{-\sqrt{z}} e^{sz} dz \\ &= - \oint_{\delta_6} e^{-\sqrt{z}} e^{sz} dz + \int_R^\delta e^{-\sqrt{z}} e^{sz} (e^{i\pi}) dx + \oint_{C_\infty} e^{-\sqrt{z}} e^{sz} dz \\ & \quad + \int_\delta^R e^{-\sqrt{z}} e^{sz} (e^{-i\pi}) dx + \int_{\delta-iT}^{\delta+iT} e^{-\sqrt{z}} e^{sz} dz \end{aligned}$$

$$\therefore \oint_C e^{-\sqrt{s}} e^{sz} dz = - \oint_{\delta_6} e^{-\sqrt{s}} e^{sz} dz + \int_0^R e^{-i\sqrt{s}} e^{-sx} dx + \oint_{C_\infty} e^{-\sqrt{s}} e^{sz} dz \\ + \int_0^R -e^{i\sqrt{s}} e^{-sx} dx + \int_{\delta-i\pi}^{\delta+i\pi} e^{-\sqrt{s}} e^{sz} dz$$

for $R \rightarrow \infty$, assume $\oint_{C_\infty} e^{-\sqrt{s}} e^{sz} dz \rightarrow 0$

for $\delta \rightarrow 0$,

$$\oint_{\delta_6} e^{-\sqrt{s}} e^{sz} dz = \int_0^{2\pi} e^{-\sqrt{s} e^{i\frac{\theta}{2}}} e^{s\delta e^{i\theta}} (i\delta e^{i\theta}) d\theta \rightarrow 0$$

for closed contour of analytic function, $\oint_C e^{-\sqrt{s}} e^{sz} dz = 0$

Therefore,

$$\int_0^\infty e^{-i\sqrt{s}} e^{-sx} dx + \int_0^\infty -e^{i\sqrt{s}} e^{-sx} dx + \int_{\delta-i\infty}^{\delta+i\infty} e^{-\sqrt{s}} e^{sz} dz = 0$$

$$\therefore \mathcal{L}^{-1}\{e^{-\sqrt{s}}\} = \frac{1}{2\pi i} \int_{\delta-i\infty}^{\delta+i\infty} e^{-\sqrt{s}} e^{sz} dz \\ = -\frac{1}{2\pi i} \int_0^\infty (e^{-i\sqrt{x}} - e^{i\sqrt{x}}) e^{-sx} dx \\ = -\frac{1}{2\pi i} \int_0^\infty -2i \sin \sqrt{x} e^{-sx} dx \\ = \frac{1}{\pi} \int_0^\infty \sin \sqrt{x} e^{-sx} dx$$

Let $x=r$, $s=t$, therefore $\mathcal{L}^{-1}\{e^{-\sqrt{s}}\} = \frac{1}{\pi} \int_0^\infty \sin \sqrt{r} e^{-rt} dr$

(b) Let $rt = u^2$, therefore $\frac{dr}{du} = \frac{2u}{t}$

$$\frac{1}{\pi} \int_0^\infty \sin \sqrt{r} e^{-rt} dr = \frac{1}{\pi} \int_0^\infty \sin\left(\frac{u}{\sqrt{t}}\right) e^{-u^2} \frac{2u}{t} du \\ = \text{Im} \left\{ \frac{1}{\pi t} \int_0^\infty u e^{-u^2} e^{i\frac{u}{\sqrt{t}}} du \right\}$$

$$(c) \oint_C w e^{-(w-\frac{1}{2\sqrt{t}})^2} dw = \int_{-R}^R u e^{-(u-\frac{1}{2\sqrt{t}})^2} du + \int_0^{\frac{1}{2\sqrt{t}}} (R+iv) e^{-(R+i(v-\frac{1}{2\sqrt{t}}))^2} dv \\ + \int_R^{-R} (u+i\frac{1}{2\sqrt{t}}) e^{-u^2} du + \int_{\frac{1}{2\sqrt{t}}}^0 (-R+iv) e^{-(-R+i(v-\frac{1}{2\sqrt{t}}))^2} dv$$

$$\text{for } R \rightarrow \infty, \int_0^{\frac{1}{2\sqrt{t}}} (R + iv) e^{-[R + i(v - \frac{1}{2\sqrt{t}})]^2} dv \rightarrow 0$$

$$\int_{\frac{1}{2\sqrt{t}}}^0 (-R + iv) e^{-[-R + i(v - \frac{1}{2\sqrt{t}})]^2} dv \rightarrow 0$$

$$\therefore \oint_C w e^{-w^2 - \frac{1}{4t}} dw = e^{\frac{1}{4t}} \int_{-\infty}^{\infty} u e^{-u^2} e^{i\frac{u}{\sqrt{t}}} du + \int_{\infty}^{-\infty} (u e^{-u^2} + \frac{1}{2\sqrt{t}} i e^{-u^2}) du$$

$$\text{Along the closed contour with analytic function, } \oint_C w e^{-w^2 - \frac{1}{4t}} dw = 0$$

Therefore,

$$\begin{aligned} e^{\frac{1}{4t}} \int_{-\infty}^{\infty} u e^{-u^2} e^{i\frac{u}{\sqrt{t}}} du &= \int_{-\infty}^{\infty} (u e^{-u^2} + \frac{1}{2\sqrt{t}} i e^{-u^2}) du \\ &= \left[-\frac{1}{2} e^{-u^2} \right]_{-\infty}^{\infty} + \frac{1}{2\sqrt{t}} i \int_{-\infty}^{\infty} e^{-u^2} du \end{aligned}$$

$$\therefore \text{Im} \left\{ \int_{-\infty}^{\infty} u e^{-u^2} e^{i\frac{u}{\sqrt{t}}} du \right\} = \frac{e^{-\frac{1}{4t}}}{2\sqrt{t}} \sqrt{\pi}$$

$$\therefore \text{Im} \left\{ \frac{1}{\sqrt{4t}} \int_{-\infty}^{\infty} u e^{-u^2} e^{i\frac{u}{\sqrt{t}}} du \right\} = \frac{e^{-\frac{1}{4t}}}{2\sqrt{t} t^{\frac{1}{4}}}$$

$$\therefore \mathcal{L}^{-1} \{ e^{-\frac{1}{4t}} \} = \frac{1}{\sqrt{\pi}} \int_0^{\infty} \sin \sqrt{r} e^{-rt} dr = \frac{e^{-\frac{1}{4t}}}{2\sqrt{t} t^{\frac{1}{4}}}$$

4.6 PR4_6

(a) By using the residue theorem and the complex inversion formula for the Laplace transform, prove that

$$\mathcal{L}^{-1} \left\{ \frac{1}{s(s^2 + 2s + 3)} \right\} = \frac{1}{3} \left(1 - e^{-t} \cos(\sqrt{2} t) - \frac{e^{-t}}{\sqrt{2}} \sin(\sqrt{2} t) \right).$$

(b) By taking a Laplace transform and using the result from part (a), solve

$$\frac{\partial^2 u}{\partial t^2} + 2 \frac{\partial u}{\partial t} + x \frac{\partial u}{\partial x} = x^3, \quad u(x, 0) = u_t(x, 0) = u(0, t) = 0.$$

(a) let $f(s) = \frac{e^{st}}{s(s^2 + 2s + 3)}$

$$\text{for } s(s^2 + 2s + 3) = 0, \quad s_0 = 0 \text{ or } -1 \pm \sqrt{2}i$$

$$\text{let } p = e^{st}, \quad q = s(s^2 + 2s + 3)$$

$$\therefore q' = (s^2 + 2s + 3) + s(2s + 2)$$

$$\therefore \text{Res}_{s_0=0} f(z) = \frac{1}{3}$$

$$\text{Res}_{s_0=-1+\sqrt{2}i} f(z) = \frac{e^{-t} e^{\sqrt{2}it}}{(-1+\sqrt{2}i)2\sqrt{2}i} = \frac{e^{-t} e^{\sqrt{2}it}}{-4-2\sqrt{2}i}$$

$$\text{Res}_{s_0=-1-\sqrt{2}i} f(z) = \frac{e^{-t} e^{-\sqrt{2}it}}{(-1-\sqrt{2}i)(-2\sqrt{2}i)} = \frac{e^{-t} e^{-\sqrt{2}it}}{-4+2\sqrt{2}i}$$

$$\begin{aligned} \therefore \mathcal{L}^{-1} \left\{ \frac{1}{s(s^2 + 2s + 3)} \right\} &= \frac{1}{3} - \frac{e^{-t} e^{\sqrt{2}it}}{4+2\sqrt{2}i} - \frac{e^{-t} e^{-\sqrt{2}it}}{4-2\sqrt{2}i} \\ &= \frac{1}{3} - \frac{(4-2\sqrt{2}i)e^{-t} e^{\sqrt{2}it}}{24} - \frac{(4+2\sqrt{2}i)e^{-t} e^{-\sqrt{2}it}}{24} \\ &= \frac{1}{3} - \frac{e^{-t}(e^{\sqrt{2}it} + e^{-\sqrt{2}it})}{6} + \frac{\sqrt{2}ie^{-t}(e^{\sqrt{2}it} - e^{-\sqrt{2}it})}{12} \\ &= \frac{1}{3} - \frac{e^{-t} \cos(\sqrt{2}t)}{3} - \frac{e^{-t} \sin(\sqrt{2}t)}{3\sqrt{2}} \\ &= \frac{1}{3} \left[1 - e^{-t} \cos(\sqrt{2}t) - \frac{e^{-t}}{\sqrt{2}} \sin(\sqrt{2}t) \right] \end{aligned}$$

$$(b) \quad s^2 U - su(x, 0) - u_t(x, 0) + 2sU - u(x, 0) + x \frac{\partial U}{\partial x} = \frac{x^3}{s}$$

$$\therefore \frac{\partial U}{\partial x} + \frac{s^2 + 2s}{x} U = \frac{x^2}{s}$$

$$\text{Let } I(x) = e^{\int \frac{s^2 + 2s}{x} dx} = x^{s^2 + 2s}$$

$$\therefore \frac{d}{dx} (U x^{s^2 + 2s}) = \frac{1}{s} x^{s^2 + 2s + 2}$$

$$\therefore U x^{s^2 + 2s} = \frac{1}{s(s^2 + 2s + 3)} x^{s^2 + 2s + 3} + C \quad \therefore U = \frac{1}{s(s^2 + 2s + 3)} x^3 + C x^{-s^2 - 2s}$$

$$\therefore u = \frac{x^3}{3} \left[1 - e^{-t} \cos(\sqrt{2}t) - \frac{e^{-t}}{\sqrt{2}} \sin(\sqrt{2}t) \right] + \mathcal{L}^{-1} \{ C x^{-s^2 - 2s} \}$$

$$\text{where } \mathcal{L}^{-1} \{ C x^{-s^2 - 2s} \} = 0 \quad \text{as } u(0, t) = 0$$

4.7 PR4_7

(a) Show that if $f(t) = \int_t^\infty \frac{g(u)}{u} du$ then $F(s) = \frac{1}{s} \int_0^s G(q) dq$.

(b) Show that the Laplace transform of $f(t) = \int_t^\infty \frac{e^{-u}}{u} du$ is $\frac{1}{s} \ln(s+1)$.

$$\begin{aligned}
 \text{(a)} \quad F(s) &= \int_0^\infty e^{-st} \int_t^\infty \frac{g(u)}{u} du dt \\
 &= \int_0^\infty e^{-st} \int_0^u \frac{g(u)}{u} dt du \\
 &= \int_0^\infty \frac{g(u)}{u} \int_0^u e^{-st} dt du \\
 &= \int_0^\infty \left(-\frac{1}{s} e^{-ut} + \frac{1}{s} \right) \frac{g(u)}{u} du \\
 &= \frac{1}{s} \int_0^\infty (1 - e^{-tu}) \frac{g(u)}{u} du
 \end{aligned}$$

$$\begin{aligned}
 \therefore \int_0^s G(q) dq &= \int_0^s \int_0^\infty e^{-qu} g(u) du dq \\
 &= \int_0^\infty \int_0^s e^{-qu} g(u) dq du \\
 &= \int_0^\infty g(u) \int_0^s e^{-qu} dq du \\
 &= \int_0^\infty (1 - e^{-su}) \frac{g(u)}{u} du
 \end{aligned}$$

$$\therefore F(s) = \frac{1}{s} \int_0^s G(q) dq$$

$$\text{(b)} \quad F(s) = \frac{1}{s} \int_0^s G(q) dq$$

$$\text{where } g(u) = e^{-u}$$

$$\therefore G(q) = \frac{1}{q+1}$$

$$\therefore F(s) = \frac{1}{s} \int_0^s G(q) dq = \frac{1}{s} \int_0^s \frac{1}{q+1} dq$$

$$\therefore F(s) = \frac{1}{s} \ln(s+1)$$